

Consolidated RTH Gap Analysis Report: NQ

Date: January 23, 2026

Ticker: NQ1 (NQ)

Data Range: 2006-01-06 to 2026-01-22 (4962 Sessions)

Script: scripts/analysis/analyze_gap_history.py

1. Executive Summary

This analysis investigates the behavior of **Regular Trading Hours (RTH) Gaps** for NQ. Key findings show that NQ gaps fill approximately 66.8% of the time.

2. Terminology: Reversion vs. Defense

Term	Strategy	Market Context	Bias Edge
Reversion Favored	Trade for the Fill. Fade the gap move back to yesterday's close.	Low ATR, Low VVIX.	High Fill Rate.
Defense Favored	Trade for Continuation. Bet on the gap holding (The 'Moat').	High ATR, High VVIX.	High Defense Rate.

3. Daily Bias Inference: Morning Checklist

Use this logic gate every morning at 09:30 ET:

STEP 1: Check the Environment

- **Volatility:** Is VVIX > 110 or is ATR High? (If yes -> **Defense Favored**).

- **News:** Is there an 8:30 AM US News release (NFP/CPI)? (If yes -> **Expect wider volatility before fill**).

STEP 2: Measure the Gap Size

- **Gap < 0.15%:** High probability **Reversion** (Treat as noise).
- **Gap 0.15% - 0.45%:** The **Conflict Zone**. Lean on Volatility/Context filters.
- **Gap > 0.45%:** High probability **Defense** (Expect Trend Continuation).

STEP 3: The 15-Minute Execution Filter

- **The Moat Check:** If Yesterday's Extreme (High/Low) holds for the first 15m, the **Defense** bias is confirmed.

4. Fill Probabilities by Size

bucket	Days	Fill Rate
Very Small (<0.07%)	1038	93.5%
Small (0.07-0.15%)	981	79.0%
Medium (0.15-0.25%)	884	68.4%
Large (0.25-0.45%)	960	54.1%
Very Large (>0.45%)	1065	38.6%

[!TIP]

Takeaway: Small gaps (<0.15%) are largely noise and revert quickly. Large gaps (>0.45%) represent true 'Signal' and have a much higher probability of defending the open.

5. Day of Week Analysis

day	Count	Fill Rate	Med Time (min)
Monday	932	60.6%	13
Tuesday	1027	68.0%	17

day	Count	Fill Rate	Med Time (min)
Wednesday	1022	70.0%	17
Thursday	1003	69.0%	16
Friday	978	66.0%	13

[!IMPORTANT]

Takeaway: Mid-week (Wednesday) typically shows highest reversion. Mondays/Fridays are 'Defense' prone.

6. Time-to-Fill Distribution (Module 1)

t_bucket	Fill Count	% of All Fills	Cumulative %
0-5m	609	18.4	18.4
5-15m	506	15.3	33.7
15-30m	382	11.5	45.2
30-60m	368	11.1	56.3
60-120m	336	10.1	66.4
120m+	567	17.1	83.5

Intensity by Gap Size

bucket	Med Fill Time	% Filled <15m	Fill Count
Very Small (<0.07%)	1	83.5	971
Small (0.07-0.15%)	11	54.8	775
Medium (0.15-0.25%)	25	36.2	605
Large (0.25-0.45%)	50	20	519
Very Large (>0.45%)	100	7.5	411

Takeaway: If a fill isn't achieved in the first 30 minutes, probability of same-day fill drops. High-conviction reversions happen fast (<15m).

7. Partial Fill Behavior (Module 2)

r_bucket	Count	% of Total
0-25%	540	10.9
25-50%	416	8.4
50-75%	349	7
75-99%	326	6.6
100%	3315	66.8

Conditional Probability of Full Fill

If Reached X%	P(Full Fill)	Sample
25%	75.2%	4410
50%	82.8%	4003
75%	90.8%	3649

Takeaway: If price 'hangs' at the 50% retracement level for more than 15m, failure probability increases.

8. Consecutive Day / Streak Analysis (Module 3)

	Fill Rate	Days
Prior Defended	69.4%	1648
Prior Filled	65.5%	3314

Streak Persistence

streak_count	Fill Rate	Sample Size
1	68.6%	1114
2	81.9%	597

streak_count	Fill Rate	Sample Size
3	90.1%	353
4	96.9%	227
5	98.6%	144

Takeaway: Volatility and outcomes cluster. After two consecutive fills, expect a defense day soon.

9. Globex Range Context (Module 4)

globex_bucket	Fill Rate	Days
Narrow (<50%)	75.8%	2123
Normal (50-100%)	62.8%	2435
Wide (>100%)	42.6%	387

RTH Open Position

globex_position	Fill Rate	Days
Lower Third	63.3%	1434
Middle Third	78.0%	1553
Upper Third	60.6%	1975

Takeaway: Wide Globex ranges signal breakaway. Position within range helps predict early defense.

10. MAE / MFE Precision (Stats Trader View)

A. The 'Fakeout' Move (MFE before Fill)

- **Median Fakeout:** 83.3% of Gap Size.
- **Mean Fakeout:** 184.3%.

Takeaway: Set stops beyond 50-80% of gap size.

B. Retracement Depth (MAE for Trend)

- **Median Retrace:** 100.0%
- **Mean Retrace:** 81.4%.

C. Total Extension (MFE for Trend)

- **Median Extension:** Mean: 608.7 | Med: 160.1 | Mode: 0.0

Takeaway: Trending gaps run 1.5x to 2x the unit of the gap.

D. Pure Price Percentage Levels

Metric	Mean
MAE (Retrace)	0.52%
MFE (Fakeout)	0.34%

11. Trend & Bias Correlation

	Fill Rate	Days
('Bearish', 'DOWN')	69.9%	857
('Bearish', 'UP')	66.5%	1017
('Bullish', 'DOWN')	69.9%	971
('Bullish', 'UP')	68.1%	1248
('Unknown', 'DOWN')	62.4%	370
('Unknown', 'UP')	56.1%	499

Takeaway: Gaps against trend revert more often.

12. 🏆 Highest Edge Setups (Compound Detector)

Setup	Fill Rate	Days	Lift
Perfect Reversion	88.9%	1085	+22.1%

Setup	Fill Rate	Days	Lift
Trend Continuation	36.9%	555	-29.9%

13. Best Practices & Operational Guardrails

- 1. **Size Filter:** Gaps 0.15% - 0.45% are optimal.
- 2. **Regime Respect:** Use caution in High VIX/VVIX regimes.
- 3. **15-Minute Moat:** Wait for RTH opening candle confirmation.

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